Homework 1 (Linear Algebra Review)

January 12, 2026

This assignment is primarily a review of linear algebra. It is due on Thursday, Jan. 22nd. at Midnight. Each question will be worth 10 points. You must show your work to receive credit.

0)

Complete the Introduction Survey on Canvas. This can be found under the "Quizzes" tab. **Please complete this by Friday, Jan. 16.**

1)

Find all values of h, k such that the following linear system has

a) A single unique solution
b) No solutions
c) Infinitely many solutions

$$\begin{aligned} 2x + 4ky &= 4 \\ 3x + 6y &= h \end{aligned}$$

2)

Find the eigenvalues and corresponding eigenvectors of the following matrix:

$$A = \begin{pmatrix} 6 & -1 & -1 \\ 11 & -2 & -1 \\ 4 & 0 & -1 \end{pmatrix}$$

3)

Prove the following properties of the matrix transpose:

Let A, B be $m \times n$ and $n \times k$ matrices respectively.

a)

$$\left(A^T\right)^T = A$$

b)

$$(AB)^T = B^T A^T$$

c)

Let $\vec{a}, \vec{b} \in \mathbb{R}^n$. Show

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

4)

Let $\vec{a}, \vec{b} \in \mathbb{R}^n$. Prove the Schwarz inequality:

$$\left|\vec{a} \cdot \vec{b}\right| \leq \|\vec{a}\| \left\|\vec{b}\right\|$$

5)

Let $T : V \to W$ be a linear transformation from vector space V to vector space W. Let H be a subspace of V. Show that $T(H) = \{T(\vec{x}) \,|\vec{x} \in H\}$ is a subspace of W.

6)

Let λ be an eigenvalue of matrix A. Show that $B = A - \lambda I$ is a singular matrix.